

How to Be a Good AI Researcher

Tao LIN

September 28, 2026



1 Recitation

2 How to Do Research

- More on How to Read Papers
- How to Come up With Research Ideas?
- How to Do Experiments?
- How to Create More Impact
- 12 Resolutions for Grad Students
- Experiment Management
- How to Manage Your Time?
- How to Be Productive?
- Tips for Work-Life Balance (WLB)
- Other Career Tips

Acknowledgement

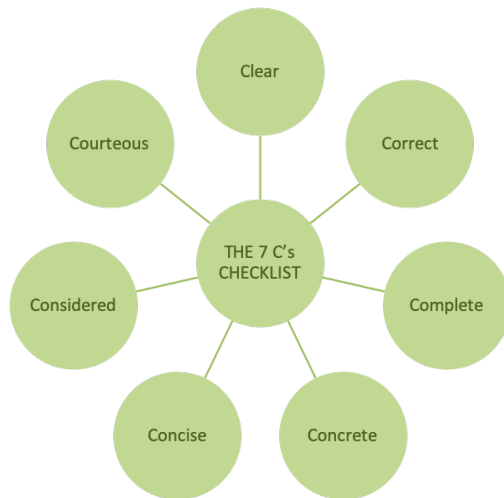
- [The 7 Cs of Communication](#), World of Work Project
- [Awesome Tips](#), JiaBin Huang
- [12 resolutions for grad students](#), Matt Might
- [Tips for work-life balance](#), Matt Might
- [Productivity tips for academics](#), Matt Might
- [How I read research papers](#), Aaditya Ramdas
- GAMES003

Table of Contents

- 1 Recitation
- 2 How to Do Research

In the previous three lectures...

We learn how to communicate: the 7 **C**'s of communication



Please revisit our previous lecture when necessary.

We learn the principles of presentation

Great talks require effort & time!

We learn the first course on “how to do research”

- The Illustrated Guide to a Ph.D.
- 10 Easy Ways to Fail a Ph.D.
- How to Make Steady Progress?
- How to Keep Track With the Literature?
- How to Read Papers?

Table of Contents

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2 How to Do Research

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But why do we need to answer these questions?

Know why you are reading a research paper

- Good reasons:
- Bad reasons:

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the original authors’ proof was complicated and has been far simplified in later works

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- “I am citing this paper, so I should read it fully.”

But verify the specific claim you cite and the conditions under which it holds.

How reading papers evolved for me (and might evolve for you)

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- Start of PhD

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I never read a paper from start to end on my first opening (or ever)

Common (wrong) belief: papers should be read linearly

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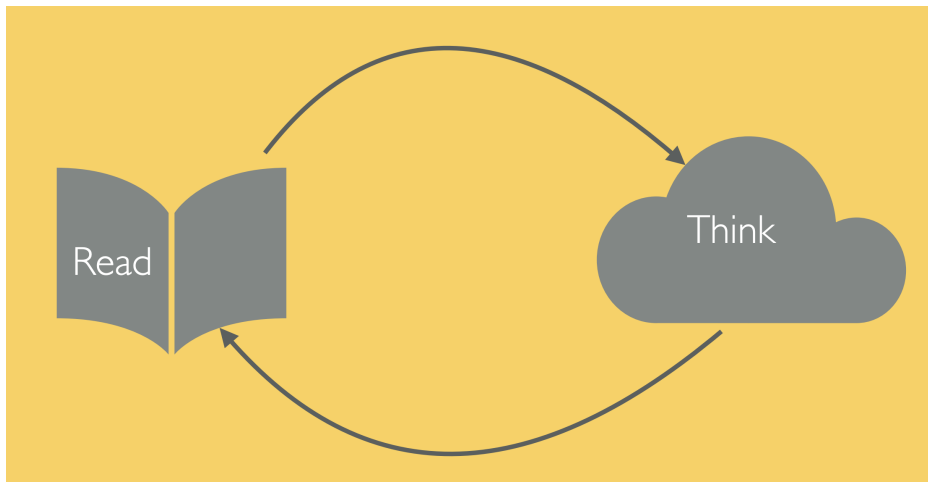
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- Sometimes the reading needs to split across days

The principle of iterative refinement



First pass: jigsaw puzzle theme (5-30mins)

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Sources: abstract/intro, problem definition, main theorem, discussion.

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Sources: examples, special cases, key lemmas/propositions, proof outlines

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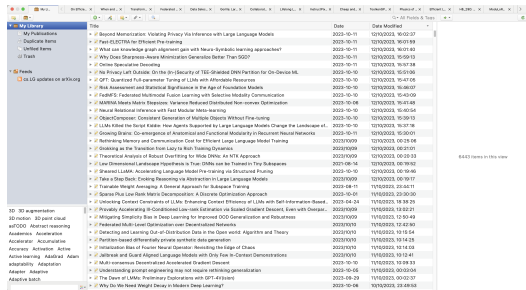
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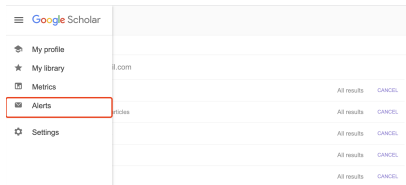
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Sources: appendices, proof details, corollaries, remarks, related work

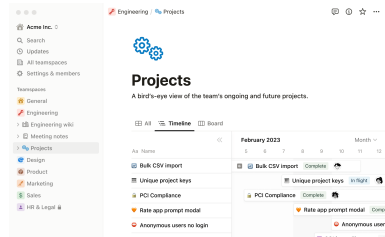
How I organize my reading



Zotero



Google Scholar Alerts



Notion

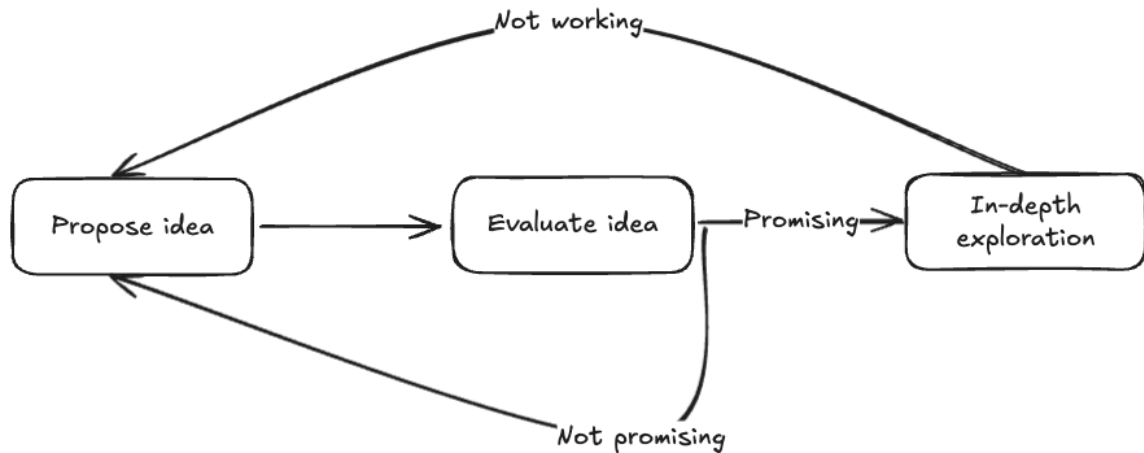
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A general loop



Manipulate assumptions

- Relax assumptions

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- Make more assumptions

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 - Identify the underlying assumptions of existing work
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 - Take a general approach and tailor it to your SPECIFIC problem.
 - Leverage all the domain knowledge (i.e., make more assumptions) to improve the method.

Combine two ideas/problems



“To steal ideas from one person is plagiarism. To steal from many is research.” — Wilson Mizner

Add an adjective

Given an existing idea X, add an adjective to make it

- slow $\rightarrow$ fast
- batch $\rightarrow$ online
- sensitive $\rightarrow$ robust
- centralized $\rightarrow$ distributed
- single-step $\rightarrow$ progressive
- single-level $\rightarrow$ hierarchical
- fixed $\rightarrow$ adaptive, sth-aware
- data-hungry $\rightarrow$ data-efficient

What makes this improvement nontrivial?

Stress test the state-of-the-art

Don't simply run on the fixed, boring benchmark datasets.

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This is where your work can fill the gap.

Building and practicing your research taste

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- 4 Read books about the history of science. Reflect on why some researchers focused on important directions their contemporaries ignored.
- 5 Critically consider your research taste, and the community taste around you. Your taste is likely very influenced by your research cluster (your collaborators, advisor, etc)

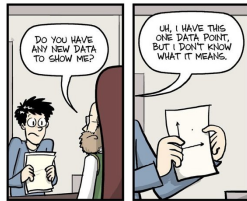
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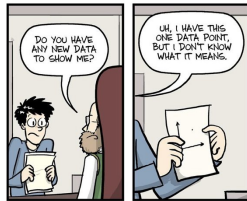
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Once we have an idea, we need to test it!



WWW.PHDCOMICS.COM

Junior students often feel stressed before the weekly meeting because their experiments do not go well.



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Some tips on why, what, and how to do experiments.

Why?

Your goal of experiments should be to validate/test your research questions.

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Your goal of experiments should be to validate/test your research questions.

- ✗ Only chase better numbers without a research question.
- ✓ Do an experiment to test a hypothesis.

What experiments should we do?

This involves three main steps:

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This involves three main steps:

- identify key research questions
- break them down into baby steps
- design experiments that best answer those questions

Example: The gain comes from module X.

Remove X; keep the training budget fixed.

Does the gain consistently disappear across repeated runs?

How?

so now we know why and what, how about HOW?

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There are no universal answers, but here are some principles I found helpful.

Some tips

- One thing at a time

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The goal of your experiment is to test a hypothesis!

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- Ask for feedback

Your mentors/advisors are there to help you succeed.

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Pick a good name

(And Pick A Good Name!)

Reduced
Instruction

Set
Computers

Redundant
Array of
Inexpensive
Disks

Network
Of
Workstations

...

30

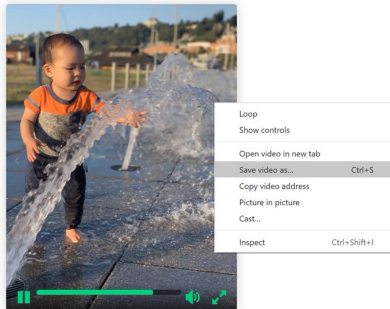


Make all your results available

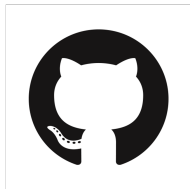
3D Photography using Context-aware Layered Depth Inpainting

Meng-Li Shih^{1,2} Shih-Yang Su¹ Johannes Kopf³ Jia-Bin Huang¹

¹Virginia Tech ²National Tsing Hua University ³Facebook



Lower the barrier for others to follow



GitHub



Google colab

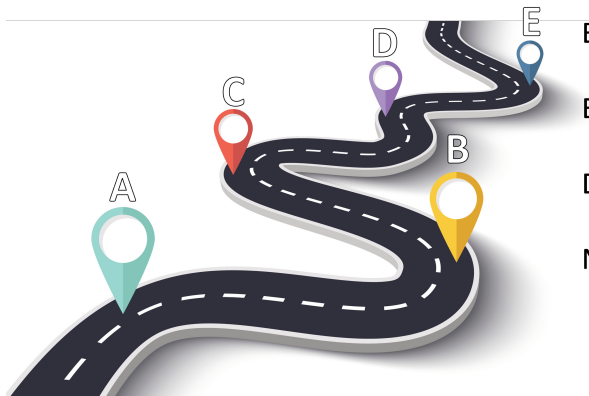


Hugging Face



PyTorch hub

Make others' life easier



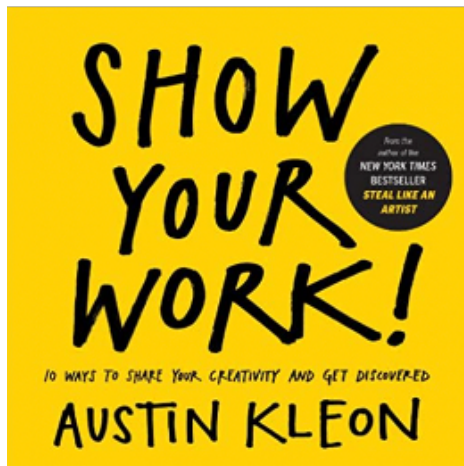
Baseline implementation?

Evaluation script?

Data preprocessing?

New dataset?

Show your work!



Your website, Zhihu, Weibo, bilibili, YouTube, Twitter, etc

This week we will talk more on ***productivity and career.***

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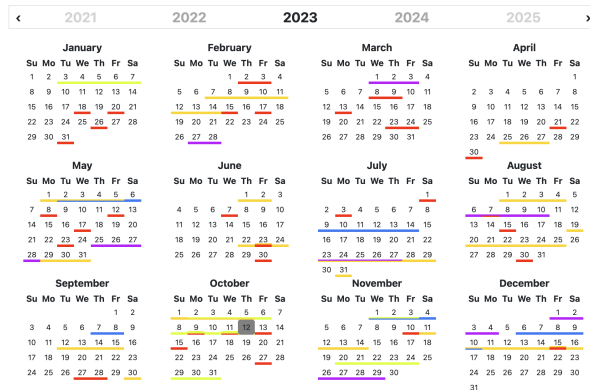
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Map out the year

Map out what the next twelve months will look like.

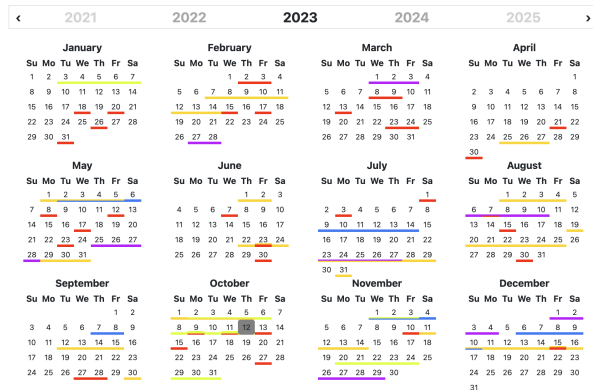
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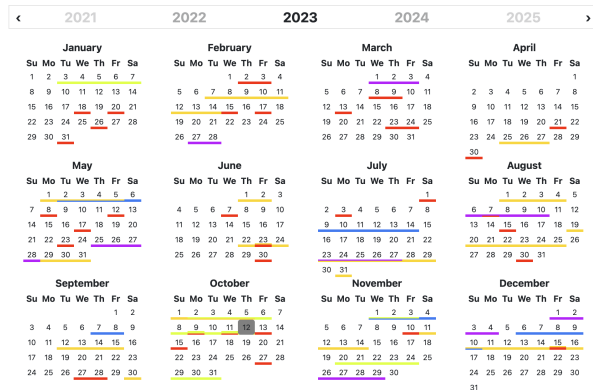
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- Put major deadlines on your calendar.

Map out the year

Map out what the next twelve months will look like.



- Put major deadlines on your calendar.
- Decide program milestones

Improve productivity

We will discuss it later!

Embrace the uncomfortable

What topics in your field are outside your comfort zone?

Embrace the uncomfortable

What topics in your field are outside your comfort zone?

Try those!

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Tips:

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Tips:

- The “rule of 3”:

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- The “15 minute rule”:

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Try those!

Tips:

- The “rule of 3”:

When a third person recommends you try something, you must try it.

- The “15 minute rule”:

Give something the benefit of the doubt for 15 minutes.

If you don't want to continue after 15 minutes, drop it.

Upgrade your tools

Make sure you have the right tools.

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Make sure you have the right tools.

For example,

Upgrade your tools

Make sure you have the right tools.

For example,

- Do you have an automatic research workflow?

Upgrade your tools

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For example,

- Do you have an automatic research workflow?
- Is there software or hardware that could accelerate your workflow?

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Upgrade your tools

Make sure you have the right tools.

For example,

- Do you have an automatic research workflow?
- Is there software or hardware that could accelerate your workflow?
- Have you optimized your configuration files?
- Is it time to set up your LaTeX macros?

Stay healthy

Mind and body are connected: a healthy body supports a creative mind.

¹You do not have to handle it alone; seeking support is encouraged.

Stay healthy

Mind and body are connected: a healthy body supports a creative mind.

- Evaluate your diet and exercise habits.

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Stay healthy

Mind and body are connected: a healthy body supports a creative mind.

- Evaluate your diet and exercise habits.
- Learn to do something with your body.

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Stay healthy

Mind and body are connected: a healthy body supports a creative mind.

- Evaluate your diet and exercise habits.
- Learn to do something with your body.
- Watch out for mental health¹.

¹ You do not have to handle it alone; seeking support is encouraged.

Update your CV and web site

Doing a Ph.D. $\approx$ building your life product.

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- For a CSer, if you can't be googled, you don't exist.

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Doing a Ph.D. $\approx$ building your life product.

- For a CSer, if you can't be googled, you don't exist.
- Maintain a well-designed, professional-looking academic web site.
- Try your best to improve your profile / product.

Keep your eye on the job market

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- Get internships at top companies for return offer.
- Look at the hiring areas and check whether your work fits.
- If your school is interviewing, attend hiring talks.

Network

Your future success will depend in part on networking.

Network

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For example,

Network

Your future success will depend in part on networking.

For example,

- You will need letter-writers.

Network

Your future success will depend in part on networking.

For example,

- You will need letter-writers.
- You will need them to pull you from the crowd when applying for jobs/interns.

Say thanks

Thank the giants upon whose shoulders you stand.

Volunteer for a talk

Effective public communication is critical to success.

Practice writing

Effective writing is equally critical to success.

Check with your committee

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- If you don't have a committee

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→ you may even uncover opportunities for collaboration.

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→ form one earlier rather than later.
- If you have a committee
→ check in with your committee at least once a year to update them on your plan.
- Check with your committee is also a way for networking
→ you may even uncover opportunities for collaboration.
- Keeping your committee informed eliminates surprises at your defense.

Table of Contents

1 Recitation

2 How to Do Research

- More on How to Read Papers
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Record each experiment:

Hypothesis / Config & code version / Result / Next step

[See an external link.](#)

Table of Contents

1 Recitation

2 How to Do Research

- More on How to Read Papers
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Why do we need good time management?

Be on top of things. Avoid drama and stress.

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It makes the finiteness of time apparent.

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Philosophy: Calendars convert time to space.



It makes the finiteness of time apparent.



Make EVERYTHING you plan to do as a calendar entry (including planning the calendar) and do it at that time.

Calendar, not to-do lists: viewing time as space².

- Principle 1: Everything takes time. So everything needs to be on your calendar.
- Principle 2: It is easier to *measure how wrong your time estimates are* than *it is to fix them*.
- Principle 3: More generally, incorporate your patterns.
- Principle 4: Re-plan.
- Principle 5: Break it down.
- Principle 6: Backtrack. Foresee.
- Principle 7: Visualize your time.

²See details in <https://deviparikh.medium.com/calendar-instead-of-to-do-lists-9ada86a512dd>

Principle 1: Everything takes time → everything on your calendar.

Time is always ticking and there is finite time in the day.

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Everything takes time
(something, anything, and nothing)

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Everything takes time

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- If you see repeatedly that you are too tired on Saturdays to do anything
 - Mark your entire Saturdays as “goof off”

Principle 4: Re-plan.

Re-planning is not failure

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- The plan is feasible. But it is not guaranteed to work.
- Move things you could not get done to other open slots.
- The future you will appreciate the extra buffer!

Principle 5: Break it down.

Tasks don't always present themselves in calendar-sized chunks.

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Tasks don't always present themselves in calendar-sized chunks.

- 1 A 300-hour project due in 3 months cannot be a calendar entry.
- 2 Break it down.
- 3 Mark them on your calendar in available slots

Principle 6: Backtrack and Foresee.

Backtrack on the timeline.

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- It means that *you need to start writing the paper at least a month before the deadline.*
- It means that *you have a month to get your main basic result.*

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→ Mark that entire week as busy.

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 - This will make sure the week before the deadline is not any more stressful than it needs to be.

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✗ Don't mistake it to be the time when you should do something.

- Re-planning needs empty slots, needs buffer.

Principle 7: Visualize your time.

Wed	Thu	Fri	Sat	Sun
7:30 AM Get ready and go to work	7:30 AM Get ready and go to work	7:30 AM Get ready and go to work	7:30 AM Sleep in	
Add LaidSiv paper to CV	PerceptiveConv meeting			Call Mom
SFA replied? Plan expenses	LaidSiv Pytorch meeting	AI Guest interview		9:30 AM Look into Atlanta apartments
Prepare for hiring meeting	Hold because talks often...	10 AM Alex Parrish's talk	10 AM Get ready, brunch, loiter	Call Shankar and Gulpreet
Write YFA grant report	Ramesh's meeting DAI biweekly	Meet RE candidate		12 PM Write time management blog post
Lunch	Lunch	Lunch		
Plan PRCV18 practice ses...	1 PM Plan for planning class in Fall	1 PM Group meeting or reading group		
David going to IACL?				
Look into Kelly's gifts data...				
2:30 PM Think about service role for next year -- let Brian know	3 PM Algorithmic art coding	Manager 1:1 When is Angela starting?	3 PM Meeting Julie and Andy	
4 PM Figure out concrete intern projects. Otherwise hard to make good progress! Set up meetings with Boris to brainstorm.		Maria's Q&A		
		5 PM Calvin, Sam transferring courses? Dawn, Mohit taking quals next semester?		
Dinner	Dinner	Dinner		Dinner
Leave \$120 for Lydia	Read article Dori sent			8 PM Black Mirror
Swati's job search status?				
Order fans				
answer question in journal	answer question in journal	answer question in journal	answer question in journal	answer question in journal
Sleep	Sleep	Sleep		Sleep

Table of Contents

1 Recitation

2 How to Do Research

- More on How to Read Papers
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When transiting from *structured learning environments* (e.g. courses) to *unstructured ones* (e.g. research)

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We need • a productive routine and • a good time management.

Exercise, exercise, exercise

Doing regular exercise (e.g., in the early morning) gives you the

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- mental acuity

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Doing regular exercise (e.g., in the early morning) gives you the

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- physical energy

Exercise, exercise, exercise

Doing regular exercise (e.g., in the early morning) gives you the

- mental acuity
- physical energy
- emotional stability

Exercise, exercise, exercise

Doing regular exercise (e.g., in the early morning) gives you the

- mental acuity
- physical energy
- emotional stability
- social health

Eat that frog

Start your day by working on the most important task of the day (aka the frog)

Stick to your plan and do what you are doing

Stick to your plan and do what you are doing

Multitasking /,mʌl.ti'tæs.kɪŋ/

(v.) Screwing up multiple things
at the same time.

Multi-tasking gives you the illusion of productivity.

Stick to your plan and do what you are doing

Multitasking /ˌmʌl.tiˈtæ.s.kɪŋ/

(v.) Screwing up multiple things
at the same time.

Multi-tasking gives you the illusion of productivity.

Follow your calendar and focus on one thing at a time instead.

Practice self-care

Take care of yourself!

Practice self-care

Take care of yourself!

- Get enough sleep,

Practice self-care

Take care of yourself!

- Get enough sleep,
- Eat well,

Practice self-care

Take care of yourself!

- Get enough sleep,
- Eat well,
- Do exercise.

Use interrupt coalescing

Group and defer interruptions

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Group and defer interruptions
(e.g., emails, slack messages, twitter feeds)
according to their urgency.

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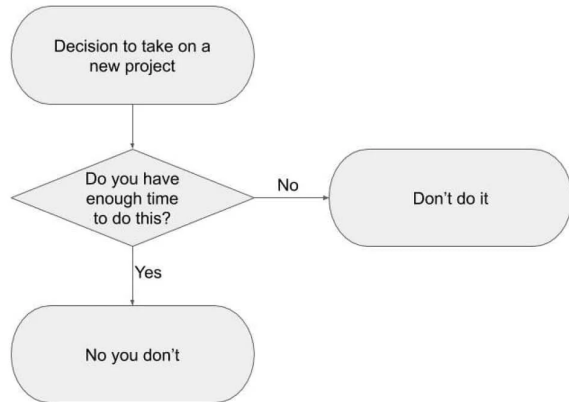
Allocate a specific time slot in a day to address these interruptions.

Learn to say no

Before accepting any new tasks, ask yourself

Learn to say no

Before accepting any new tasks, ask yourself



Give up precise task prioritization

Sometimes you may be *spending more time ranking the tasks than actually doing it.*

Touch each email/message exactly once

Open an email, skim through it, and make a quick decision
(reply/archive/allocate a time slot).

Don't work from home (?)

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Home is full of distractions

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Academics have flexible schedules

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Home is full of distractions $\longleftrightarrow$ Academics have flexible schedules

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Invest in making your work-space a comfortable, productive, enjoyable place to be, e.g.,

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Invest in making your work-space a comfortable, productive, enjoyable place to be, e.g.,

- Move your books into your work-space.

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Invest in making your work-space a comfortable, productive, enjoyable place to be, e.g.,

- Move your books into your work-space.
- Get an ergonomic office chair.

Don't work from home (?)

Home is full of distractions $\longleftrightarrow$ Academics have flexible schedules

Invest in making your work-space a comfortable, productive, enjoyable place to be, e.g.,

- Move your books into your work-space.
- Get an ergonomic office chair.
- Get a high-quality ergonomic keyboard.

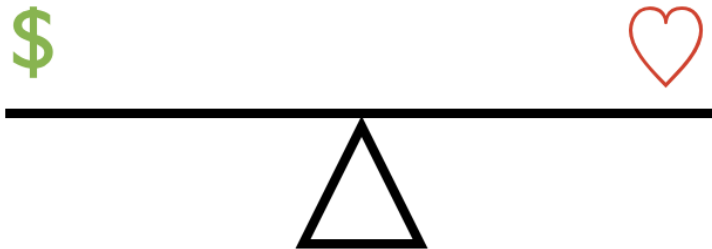
Table of Contents

1 Recitation

2 How to Do Research

- More on How to Read Papers
- How to Come up With Research Ideas?
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Tips for work-life balance



Some high-level tips³:

- Explicitly acknowledge priorities
- Avoid workaholism
- Avoid perfectionism
- Set and enforce boundaries
- Avoid over-commitment
- Use a work-flow system
- Communicate
- Keep hobbies
- Exploit opportunity cost
- Continuously adapt

³See details in <https://matt.might.net/articles/work-life-balance/>

Explicitly acknowledge priorities

Start by listing your priorities in life.

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Start by listing your priorities in life.

- ✗ If factors like “health”, “happiness”, or “family” rank below professional goals
⇒ your long-term priorities are not stable.
- ✗ If those inverted priorities persist
⇒ reconsider how your work commitments affect your well-being and relationships.

Avoid workaholism

✗ The “work more; sleep less” mentality is misguided.

Avoid workaholism

- ✗ The “work more; sleep less” mentality is misguided.
- ✓ Efficiency.

Avoid workaholism

- ✗ The “work more; sleep less” mentality is misguided.
- ✓ Efficiency.

The equation for work:

$$\text{output} = \underset{:=\text{productivity}}{\text{unit of work / hour}} \times \text{hours worked} \quad (1)$$

Avoid perfectionism

It is good to use perfection as a guide, but recognize that it is unattainable.

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Perfection is a process; not a destination.

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Iterate toward perfection!

Set and enforce boundaries

Maintaining work-life balance requires setting boundaries.

Set and enforce boundaries

Maintaining work-life balance requires setting boundaries.

For example,

- I always schedule recurring “life” time into my calendar

Avoid over-commitment

Learn when and how to say “no.”

Keep hobbies

Even the best jobs create stress.

Keep hobbies

Even the best jobs create stress.

Hobbies are a way of letting stress go before it explodes.

Table of Contents

1 Recitation

2 How to Do Research

- More on How to Read Papers
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Check out links below:

- [How to Get a Great Letter of Recommendation](#)
- [Academic job search advice](#)
- [Fantastic Faculty Jobs and How to Get Them?](#)
- See more at <https://github.com/jbhuang0604/awesome-tips#career>